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## Sharpening device


### Abstract

A device for sharpening bladed tools comprising a body defining an elongate handle portion adapted to be gripped in one hand by a user, the body having a forwardly projecting support member curved or angled downwardly with respect to the handle portion and fitted with a sharpening element at or near the end thereof, wherein the support member comprises a pair of spaced support arms extending generally parallel to the longitudinal extent of the handle portion, the support arms being positioned to respective sides of the sharpening element and defining an aperture therebetween adapted to accommodate a blade or anvil of a tool during sharpening of a counterpart tool blade.



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## Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase of International Application No. PCT/AU2020/050960 filed on Sep. 10, 2020 which claims the benefit of Australian Patent Application No. 2019903344 filed Sep. 10, 2019 entitled “Sharpening Device”, each of which is incorporated by reference herein in its entirety.

(2) This invention relates to a device for sharpening bladed instruments including gardening implements and the like.

(3) Gardening implements with blades used for cutting, such as loppers, pruners, secateurs, knives, machetes, hoes, spades, hatchets, axes, etc., should be maintained with sharp cutting edges for safe and efficient operation. Maintaining a sharp edge on the blade(s) can present a challenge for a number of reasons.

(4) The cutting edge(s) of a gardening implement can be difficult to access due to the construction of the implement. For example, secateurs are scissor-like implements having a pair of pivoted handles and may have a single cutting blade that closes against a flat surface (anvil-type) or two cutting blades (bypass-type). For ease of operation the handles of the secateurs are usually sprung so that they are normally open, but this means the blades, or blade and anvil, are limited in their separation thereby limiting access by a sharpening tool. Moreover, oftentimes the blades, or blade and anvil, are curved, which further restricts access to the edge(s) for sharpening, particularly the base of the blade.

(5) Not all garden cutting implements are held in one hand like secateurs. For instance, loppers may have a similar action but with long handles that are operated with both hands for extra leverage. The long handles make them unwieldy to manipulate. Accordingly, a sharpening device that can be used on a variety of different kinds of implement blades should be able to be safely and securely operated with one hand whilst the other hand holds steady the implement to be sharpened.

(6) As mentioned above, there are cutting tools that employ two common styles of cutting action—anvil-type and bypass-type. Anvil-type secateurs work similar to a knife where a blade is pushed through the plant material onto a cutting board, i.e. the anvil. Anvil-type secateurs generally have a blade with bevels on each side of the blade edge. Bypass-type secateurs work more like scissors where two blades pass by each other, and as a result generally have blades with one flat side and one bevelled side. For optimal sharpening, different sharpening element configurations are required for use on single-bevel and double-bevel blade types. It is known to provide sharpening tools with different sharpening elements on different parts of the tool, however this compromises the tool ergonomics as well as the effective and efficient action of the tool on the blade edge.

(7) Embodiments of the present invention aim to address the issues outlined above, or at least provide a useful alternative.

(8) In accordance with the present invention there is provided a device for sharpening bladed tools comprising a body defining an elongate handle portion adapted to be gripped in one hand by a user, the body having a forwardly projecting support member curved or angled downwardly with respect to the handle portion and fitted with a sharpening element at or near the end thereof, wherein the support member comprises a pair of spaced support arms extending generally parallel to the longitudinal extent of the handle portion, the support arms being positioned to respective sides of the sharpening element and defining an aperture therebetween adapted to accommodate a blade or anvil of a tool during sharpening of a counterpart tool blade.

(9) The sharpening element may be provided in a cartridge that is removeably fitted to the end of the support member. The sharpening element may comprise a one or more sharpening blades formed from tungsten carbide or the like. The device may be provided with two or more cartridges carrying sharpening blades configured for sharpening respective types of tool blade edges. The device body may have a compartment therein for storing one or more cartridges.

(10) In accordance with the present invention there is also provided a device for sharpening bladed

tools comprising a body defining an elongate handle portion adapted to be gripped in one hand by a user, the body having a forwardly projecting support member curved or angled downwardly with respect to the handle portion, the end of the support member being configured to engage with a sharpening element cartridge having a sharpening element held therein, wherein the support member comprises a pair of spaced support arms extending generally parallel to the longitudinal extent of the handle portion, the support arms being positioned to respective sides of the sharpening element of the engaged cartridge and defining an aperture therebetween adapted to accommodate a blade or anvil of a tool during sharpening of a counterpart tool blade.
(11) The present invention also provides a device for sharpening bladed tools comprising a body defining an elongate handle portion adapted to be gripped in one hand by a user, the body having a forwardly projecting support member curved or angled downwardly with respect to the handle portion, the end of the support member being configured to engage a first sharpening element cartridge having a first sharpening element configuration for operative tool sharpening use, the device further comprising a storage receptacle formed in the body and configured to receive a second sharpening element cartridge having a second sharpening element configuration for storage, and wherein the first and second sharpening element cartridges are selectively interchangeable by the user for operative sharpening use on different types of tool blades.
(12) In embodiments, the device includes a storage carriage adapted to engage with the sharpening element cartridge that is not, in use, fitted to the support member, the storage carriage being adapted to engage with the device body while the sharpening element cartridge carried thereby is received in the storage receptacle.
(13) The device according to embodiments of the invention may include more than two sharpening element cartridges, with each individual sharpening element cartridge having a different sharpening element configuration adapted for sharpening a respective form of bladed tool. For example, cartridges with different sharpening element configurations may be provided for use on bypass-type loppers/secateurs, anvil-type loppers/secateurs, knives/machetes/hoes/spades, hatchets/axes, etc. The storage carriage may be arranged to hold more than one cartridge at a time for storage in the receptacle.

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## Description

(1) Further aspects, features and advantages of the present invention may become apparent from the following description of an embodiment thereof, presented by way of example only, and with reference to the accompanying drawings, in which:
(2) FIG. **1** is a side view of a sharpening device according to an embodiment of the invention;
(3) FIGS. **2** and **3** are bottom and top view of the sharpening device, respectively;
(4) FIG. **4** is a rear side perspective view of the sharpening device;
(5) FIG. **5** is an upper front perspective view of the sharpening device;
(6) FIG. **6** is a lower front perspective view of the sharpening device;
(7) FIG. **7** is a side view of the sharpening device with sharpening element cartridge detached;
(8) FIG. **8** is an upper front perspective view of the sharpening device with sharpening element cartridge detached;
(9) FIG. **9** is an enlarged front perspective view of the sharpening device with sharpening element cartridge detached;
(10) FIG. **10** is an enlarged rear perspective view of the sharpening device with sharpening element cartridge detached;
(11) FIG. **11** is a side view of the sharpening device with sharpening element cartridge and storage carriage detached;
(12) FIG. **12** is a rear side perspective view of the sharpening device with sharpening element

cartridge and storage carriage detached;
(13) FIG. **13** is an enlarged rear perspective view of a portion of the sharpening device with storage carriage removed;
(14) FIG. **14** is a forward perspective view of the storage carriage fitted with a sharpening element cartridge;
(15) FIG. **15** is a front side view of the storage carriage fitted with a sharpening element cartridge;
(16) FIG. **16** is a front side view of the storage carriage with sharpening element cartridge detached;
(17) FIG. **17** is a rearward perspective view of the storage carriage with sharpening element cartridge detached;
(18) FIG. **18** is an upper rear side perspective view of the storage carriage with sharpening element cartridge detached;
(19) FIG. **19** is an upper rear side perspective view of the storage carriage in isolation;
(20) FIG. **20** is a rearward perspective view of a sharpening element cartridge;
(21) FIG. **21** is a top view of the sharpening element cartridge;
(22) FIG. **22** is a bottom view of the sharpening element cartridge;
(23) FIG. **23** a forward underside view of the sharpening element cartridge;
(24) FIGS. **24-26** are side views illustrating use of the sharpening device on a gardening implement;
(25) FIGS. **27-29** are perspective views illustrating use of the sharpening device on a gardening implement; and
(26) FIG. **30** is an exploded perspective view of a sharpening device according to another embodiment of the invention.
(27) A sharpening device **10**, **10′** constructed according to embodiments of the present invention are illustrated in various views in the drawings and described hereinbelow.
(28) The sharpening device **10** has a tool body **100** including a handle portion **110** with a forwardly projecting support member **150**. The handle portion **110** is configured to be comfortably gripped by one hand, and has its lower surface encircled by a finger guard **120**. The handle portion is elongate, with rounded and contoured upper and lower surfaces for user comfort and grip. The handle portion (or sections thereof) may comprise a “soft touch” coating, manufactured of an elastomer or other suitable polymer, for increasing user comfort and grip. The forward upper surface of the handle portion **110**, adjacent connection to the support member **150**, is provided with a thumb location depression **112**. During use, described further below, the operator grips the handle portion **110** in one hand with fingers extending between the handle and finger guard **120** and thumb aligned with the longitudinal extent of the tool body, thumb pad resting in the depression **112**.
(29) The support member **150** projects forwardly from the handle portion and, from a side view such as seem in FIG. **1**, is curved downward and tapers in to a cartridge support housing **160** provided at the end thereof. The cartridge support housing **160** has a receptacle configured to receive a sharpening element cartridge **200**. The support member is formed from right and left support arms **152**, **153** that are generally flat, plate-like structures parallel to the sides of the handle portion **110**. The right and left support arms **152**, **153** are attached to the front of the handle portion **110** and joined at their forward end by the cartridge support housing **160**, providing an aperture **155** therebetween, seen best in FIGS. **2**, **3** and **5**.
(30) At the rear of the sharpening device **10**, the tool body **100** contains a storage carriage **300**, described in detail hereinbelow. The storage carriage **300** has a butt portion **302** that forms the rear of the device **10** when in place. Below the storage carriage location the tool body is provided with a web **121** extending between the handle portion **110** and the finger guard **112**. The web **121** has a hole **122** therein which allows for attachment of a cord or the like for hanging or otherwise securing the device **10**.
(31) When fitted in the receptacle of the cartridge support housing **160**, the sharpening element

cartridge **200** forms the forward end of the sharpening device **10**, which is angled down by about sixty degrees or so by virtue of the support member **150** being curved downwardly. The sharpening element cartridge **200** has a flange **210** with a forward tapered edge, the surface of the flange **210** being substantially flush with the surface of the cartridge support housing **160** where they meet. The flange **210** has a generally V-shaped slot **202** centrally formed therein, the direction of the slot **202** being parallel to the longitudinal extent of the tool body **100**. The cartridge **200** is fitted with sharpening elements **280** firmly secured therein, portions of which extend into the slot **202**.

(32) When in use the sharpening element cartridge **200** is secured in the receptacle defined by the cartridge support housing **160**. However as seen in the drawings, for example FIGS. **7** to **10**, the sharpening element cartridge **200** is able to be detached and removed from the tool body **100**. This is enabled by latch arms **220** provided on the sides of the cartridge **200**, attached to the cartridge at the front thereof and extending parallel to the sides of the cartridge, separated from the cartridge body **205** by a gap **206**. The latch arms **220** each have a respective latch formation at the end thereof, comprising a sloped surface **222** leading to a ledge **223**. Intermediately located on each latch arm **220** is a respective release button **225** protruding outwardly to the side. When the sharpening element cartridge is received in the support housing receptacle, the latch formations are located in apertures **163** on the sides of the support housing and the release buttons **225** are received in slots **165**, which arrangement holds the cartridge in place. To remove the sharpening element cartridge from the device, the user presses inwardly on the release buttons **225** which causes the latch arms to flex inwardly so that the ledges **223** clear the apertures **163**. This permits the sharpening element cartridge **200** to be slid out of the support housing. Location of the sharpening element cartridge **200** in the cartridge support housing is further assisted by a ridge **215** formed on the rear of the cartridge, which slides into a corresponding groove inside the support housing receptacle (not seen in the drawings).

(33) The primary purpose of configuring the sharpening element cartridge **200** so that it may be removed from the cartridge support housing **160** is to allow for a different cartridge to be installed in replacement. This permits the sharpening device **10** to be selectively fitted with one of a variety of cartridges, for example a cartridge (**200**A) adapted for sharpening an anvil-type blade or a cartridge (**200**B) adapted for sharpening a bypass-type blade. The two different sharpening element cartridges **200**A, **200**B may be supplied with the device, and the cartridge not in use can be stored within the tool body **100**, as explained below.

(34) As seen best in FIGS. **12** and **13**, the handle portion **110** of the device has a hollow barrel that is open to the rear. The storage carriage **300** is adapted to fit within the hollow barrel at the rear of the tool body, and is also configured to hold the sharpening element cartridge that is not operatively fitted to front of the device at a given time.

(35) The storage carriage **300** has an elongate body **310** attached to the butt portion **302**. The carriage body **310** has an external profile of a size and shape adapted to fit relatively closely inside the hollow barrel of the tool body handle portion **110**. The forward portion of the carriage body is formed with a wall structure **340** open to one direction, the wall structure being configured to receive a sharpening element cartridge **200** when inserted from the open side. When so inserted, the sharpening element cartridge **200** is securely held by the confines of the wall structure **340**, and by the latch formation on the ends of the latch arms **220** which engage in openings **345** of the wall structure provided for that purpose. One wall of the structure **340** has a locating groove **346** that operatively receives the ridge **215** of the cartridge. The sharpening element cartridge **200** can be released and removed from the carriage **300** by pressing on the release buttons **225** of the latch arms **220**, which allows the cartridge to be pulled out.

(36) The elongate body **310** of the storage carriage **300** has on each side a latch formation **330**, just forward of the butt portion **302**. When the elongate body **310** of the storage carriage is inserted into the hollow barrel until the flange of the butt portion is adjacent the rear edge of the tool body, outward protrusions on ends of the latch formations engage in respective openings **130** on sides of

the handle portion. The protrusions on the latch formations **330** are preferably rounded to as to engage with a snap fit into the openings **130**, but allow the storage carriage to be released and removed by pulling on the butt portion **302** without having to separately press in on the latch formations **330**.

(37) By selectively placing one of the two sharpening element cartridges **200** in the cartridge support housing **160** for operative use, and the other in the storage carriage **300**, the sharpening device can be selectively configured for use on different types of tool blade edges. Once configured for the desired type of tool blade, the device can be used to sharpen the tool blade as diagrammatically illustrated in FIGS. **24-26**. These drawings show, step-wise, movement of the sharpening device along the edge of a blade of a bypass secateurs tool **500**. FIGS. **27-29** illustrate the same steps in perspective view from the opposite side.

(38) The secateurs tool **500** as seen in the figures comprises two main components pivotally coupled to one another to allow a scissor action. One component has a first handle **510** and a cutting blade **512**, the other component has a second handle **520** and a bypass blade **522**. Although it may not be readily apparent in the figures, the structure of the tool **500** prevents the handles and blades from pivoting open any further than shown, perhaps even to a lesser extent. One of the reasons for this is that the tool normally includes a spring (not shown) that biases the handles/blades into an open configuration. Some form of limiting structure is then provided to limit the separation of the handles in the open configuration to one that can be grasped by the hand.

(39) Sharpening of the tool cutting blade **512** using the sharpening device **10** involves drawing the sharpening elements **280** along the length of the blade edge **515** several times in a controlled manner. First, the forward end of the sharpening device carrying the sharpening element cartridge **200** is inserted into the space between the tool blades **512**, **522** such that the cutting blade edge **515** rests in the slot **202** containing the sharpening elements. The end of the sharpening device **10** is initially located as close as possible to the juncture where the tool blades **512**, **522** meet (FIGS. **24**, **27**). For optimal results the sharpening elements **280** should be oriented substantially orthogonal in relation to the cutting blade while allowing the user to apply some pressure. This arrangement is enabled by the structure of the support member **150**, being angled downwardly in relation to the handle and having a central aperture **155** between the support arms **152**, **153** that is able to accommodate the bypass blade **522**, as seen particularly in FIGS. **24** and **27**.

(40) To sharpen the blade **512** the user will typically hold the tool first handle **510** in one hand and the sharpening device **10** in the other hand, drawing the sharpening elements **280** in the groove **202** along the length of the cutting blade edge **515**, as shown in sequence in FIGS. **24-26** and **27-29**. Over the length of the stroke, the handle of the sharpening device **10** should remain aligned with the blade edge, and the sharpening elements should remain substantially perpendicular to the portion of the blade edge with which it is in contact.

(41) Although the illustrated embodiment of the sharpening device **10** described above includes only two different sharpening element cartridges **200** it is equally possible to provide more than two, with each individual sharpening element cartridge having a different sharpening element configuration adapted for sharpening a respective form of bladed tool. For example, cartridges with different sharpening element configurations (e.g. different angle arrangements of the sharpening elements) may be provided for use on bypass-type loppers/secateurs, anvil-type loppers/secateurs, knives/machetes/hoes/spades, hatchets/axes, etc. Yet other examples of cartridges comprise elements that are auxiliary to the sharpening process, for example, a deburring cartridge that includes a ceramic element for deburring a blade after it has been sharpened.

(42) FIG. **30** illustrates another embodiment of the sharpening device **10′** that is configured to hold more than one cartridge at a time for storage in the receptacle. The carriage body **310′** is longer than the carriage body **310** of the preceding example, in order to receive more than one cartridge (in this example, the wall structure **340′** of carriage body **310′** can hold up to five cartridges). For ease of identification by the user, the individual cartridges adapted for different sharpening

applications may be manufactured in different colours that are readily associated with the tools for which they are intended.
(43) The embodiment **10**′ of FIG. **30** also illustrates another example of the mechanism for attaching the storage carriage **300**′ to the body **110** of the sharpener. The elongate body **310**′ of the storage carriage **300**′ has a latch **600** attached to a finger grip portion **610**, just forward of the butt portion **302**′. When the elongate body **310**′ of the storage carriage is inserted into the hollow barrel, the outward protrusion on the end of the latch **600** engages with a ridge **620** formed on the inside of the wall of the handle portion. The latch **600** elastically deflects as the storage carriage **300**′ is inserted into the body **110** so that the latch **600** snap fits to the ridge **620**, locking the carriage **300**′ in place. To release the carriage **300**′, the user presses on the finger grip portion **610** to deflect the latch **600** and disengage it from the ridge **620**.
(44) The invention has been described by way of non-limiting example only and many modifications and variations may be made thereto without departing from the spirit and scope of the invention.
(45) Throughout this specification and the claims which follow, unless the context requires otherwise, the word “comprise”, and variations such as “comprises” and “comprising”, will be understood to imply the inclusion of a stated integer or step or group of integers or steps but not the exclusion of any other integer or step or group of integers or steps.
(46) The following is a list of parts referred to in the described embodiments with corresponding reference numerals:
(47) TABLE-US-00001 10, 10′ sharpening device 100 tool body 110 handle portion 112 thumb location depression 120 finger guard 121 handle web 122 attachment hole 130 carriage latch openings 150 support member 152 right support arm 153 left support arm 155 support member aperture 160 cartridge support housing 163 latch apertures 165 release button slots 200 sharpening element cartridge .sup.  200A anvil-type sharpening cartridge .sup.  200B bypass-type sharpening cartridge 202 V-shaped slot 205 cartridge body 206 latch arm gap 210 cartridge flange 215 cartridge body ridge 220 cartridge latch arms 222 sloped surface 223 ledge 225 release button 280 sharpening elements 300, 300′ storage carriage 302, 302′ storage carriage butt portion 310, 310′ carriage body 330 carriage latch formations 340, 340′ carriage wall structure 345 latch openings 346 locating groove 500 secateurs tool 510 first handle 512 cutting blade 515 cutting blade edge 520 second handle 522 bypass blade 600 latch 610 finger grip portion 620 ridge

## Claims

1. A device for sharpening bladed tools comprising: a body defining an elongate handle portion adapted to be gripped in one hand by a user, the body having a forwardly projecting support member curved or angled with respect to the handle portion, an end of the support member being configured to engage one of a plurality of sharpening element cartridges for operative use thereby defining an operative cartridge engagement position, the plurality of sharpening element cartridges each having a sharpening element, the body of the device also including a storage receptacle formed in the body and configured to receive at least one of the plurality of sharpening element cartridges for storage; and a carriage member adapted to receive and releasably engage with one or more of the plurality of sharpening element cartridges when not, in use, in the operative cartridge engagement position, the carriage member being adapted to engage with the device body while the one or more sharpening element cartridge carried thereby is received in the storage receptacle; wherein the plurality of sharpening element cartridges are each selectively interchangeable with one another by the user as between the operative engagement position and the storage receptacle.
2. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges has a distinct sharpening element configuration adapted for sharpening different tool blade types.

3. The device according to claim 1 further comprising: the plurality of sharpening element cartridges; and wherein each of the plurality of sharpening element cartridges is provided with engaging features designed for snap fit or frictional engagement with features of the carriage member to, in use, secure the cartridge while stowed in the storage receptacle.
4. The device according to claim 3, wherein the storage receptacle comprises an interior cavity formed in the handle portion, into which the carriage member can slide.
5. The device according to claim 3, wherein the carriage member can hold two or more cartridges for storage.
6. The device according to claim 3, wherein the carriage member can hold at least five cartridges for storage.
7. The device according to claim 1, wherein the end of the support member is provided with a cartridge support housing defining the operative cartridge engagement position.
8. The device according to claim 7, wherein the support member comprises a pair of spaced support arms extending generally parallel to the longitudinal extent of the handle portion, the support arms terminating at respective sides of the cartridge support housing, wherein an aperture between the support arms serves to accommodate a blade or anvil of a tool during sharpening of a counterpart tool blade.
9. The device according to claim 7 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges is provided with alignment features that in use interfit with features of the cartridge support housing to ensure correct alignment of the cartridge when in the operative engagement position.
10. The device according to claim 9 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges is provided with engaging features designed for snap fit or frictional engagement with features of the cartridge support housing to in use secure the cartridge at the operative cartridge engagement position.
11. A device for sharpening bladed tools comprising: a plurality of sharpening element cartridges each having a sharpening element; a body defining an elongate handle portion adapted to be gripped in one hand by a user, the body having a support member projecting forwardly from the handle portion to an end having an operative cartridge engagement formation engaging with one of the plurality of sharpening element cartridges in an operative cartridge engagement position, wherein the support member comprises a pair of spaced support arms extending generally parallel to the longitudinal extent of the handle portion, the support arms being positioned to respective sides of the cartridge engagement formation and defining an aperture therebetween adapted to accommodate a blade or anvil of a tool during sharpening of a counterpart tool blade, the body of the device further including a storage receptacle; and a carriage member adapted to receive and releasably engage with the one or more sharpening element cartridges that are not, in use, in the operative cartridge engagement position, the carriage member being adapted to engage with the device body while the one or more sharpening element cartridge carried thereby is received in the storage receptacle; wherein the device includes at least first and second sharpening element cartridges each of which is selectively interchangeable with one another and adapted to engage with the operative cartridge engagement formation or be placed in the storage receptacle according to user preference.
12. The device according to claim 11, wherein each of the plurality of sharpening element cartridges is provided with engaging features designed for snap fit or frictional engagement with features of the carriage member to in use secure the cartridge while stowed in the storage receptacle.
13. The device according to claim 11, wherein the storage receptacle comprises an interior cavity formed in the handle portion, into which the carriage member can slide.
14. The device according to claim 11, wherein the carriage member can hold two or more cartridges for storage.

15. The device according to claim 11, wherein the carriage member can hold at least five cartridges for storage.
16. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges is in an exposed position and accessible for sharpening when engaged with the support member in the operative engagement position.
17. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges is in a concealed position and inaccessible for sharpening when stored in the storage receptacle.
18. The device according to claim 11, wherein each of the plurality of sharpening element cartridges is in an exposed position and accessible for sharpening when engaged with the support member in the operative engagement position.
19. The device according to claim 11, wherein each of the plurality of sharpening element cartridges is in a concealed position and inaccessible for sharpening when stored in the storage receptacle.
20. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein the plurality of sharpening element cartridges are identical in structure to be selectively interchangeable with one another, with each sharpening element cartridge having a different sharpening element configuration adapted for sharpening a respective form of bladed tool.
21. The device according to claim 11, wherein the plurality of sharpening element cartridges are identical in structure to be selectively interchangeable with one another, with each sharpening element cartridge having a different sharpening element configuration adapted for sharpening a respective form of bladed tool.
22. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges has a flange with a forward tapered edge and a V-shaped slot centrally formed therein, the surface of the flange being substantially flush with the surface of the support member end where they meet.
23. The device according to claim 11, wherein each of the plurality of sharpening element cartridges has a flange with a forward tapered edge and a V-shaped slot centrally formed therein, the surface of the flange being substantially flush with the surface of the support member end where they meet.
24. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein each of the plurality of sharpening element cartridges is in a position in the storage receptacle that is 90 degrees to its position in the operative engagement position.
25. The device according to claim 11, wherein each of the plurality of sharpening element cartridges is in a position in the storage receptacle that is 90 degrees to its position in the operative engagement position.
26. The device according to claim 1, wherein the handle portion is a closed ring.
27. The device according to claim 11, wherein the handle portion is a closed ring.
28. The device according to claim 1 further comprising: the plurality of sharpening element cartridges, wherein when one of the plurality of sharpening element cartridges is coupled to the support member, the remainder of the plurality of sharpening element cartridges are coupled to the storage receptacle.